

Absolute Value Functions

A **family of functions** is a group of functions with the same fundamental characteristics.

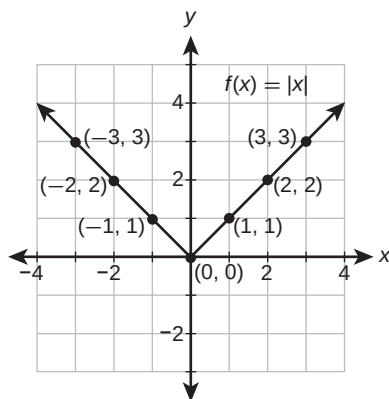
Definition

An **absolute value function** is a function whose rule contains an absolute value expression.

► **Remember** The absolute value of a number is its distance from zero on the number line.

The basic function in a family of functions is called the **parent function**. The parent function of the absolute value family of functions is $f(x) = |x|$. A table of ordered pairs and the graph of the function are shown.

x	$ x $	$f(x)$
-3	$ -3 $	3
-2	$ -2 $	2
-1	$ -1 $	1
0	$ 0 $	0
1	$ 1 $	1
2	$ 2 $	2
3	$ 3 $	3



The graph is made up of two pieces—two rays that extend from a common vertex, forming a V-shape. All absolute value functions are V-shaped. Their vertex position, orientation, and width, however, can vary.

Graphing Absolute Value Functions by Plotting Points

EXAMPLE 1

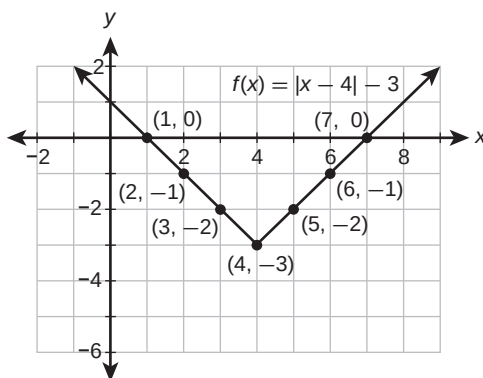
Graph $f(x) = |x - 4| - 3$.

► **Think About It** Choose x -values that allow you to determine the position of the vertex as well as both rays.

SOLUTION

Make a table of values and plot the ordered pairs.

x	$ x - 4 - 3$	$f(x)$
1	$ 1 - 4 - 3$	0
2	$ 2 - 4 - 3$	-1
3	$ 3 - 4 - 3$	-2
4	$ 4 - 4 - 3$	-3
5	$ 5 - 4 - 3$	-2
6	$ 6 - 4 - 3$	-1
7	$ 7 - 4 - 3$	0



Translating Absolute Value Functions

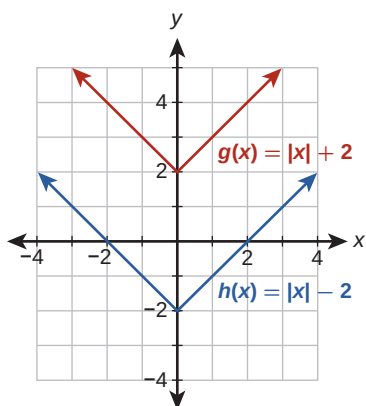
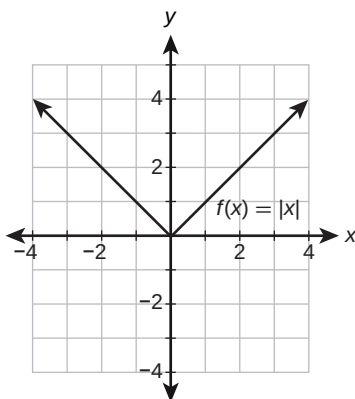
Graphing an absolute value function by plotting points can take some guess-and-check work to determine which values in the domain you should use. In a family of functions, however, once you know the graph of the parent function, you can use transformations to graph related functions.

Recall from geometry that a **transformation** is a one-to-one mapping between two sets of points. One type of transformation is a **translation**, or slide. You can translate a function horizontally, vertically, or both.

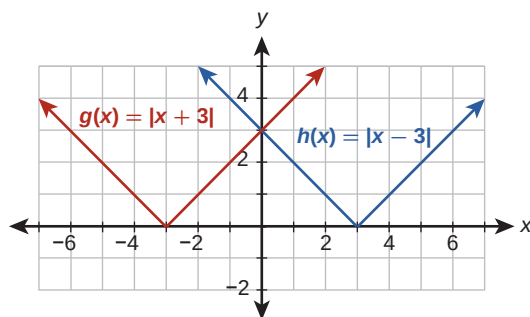
Graphs of Absolute Value Functions

Absolute Value Function Graph Family: $f(x) = |x - h| + k$

When $f(x) = |x|$ is translated k units vertically and h units horizontally, the function becomes $g(x) = |x - h| + k$. The vertex shifts from $(0, 0)$ to (h, k) .



If k is positive, then the vertex is shifted up. If k is negative, then the vertex is shifted down.



If h is positive, then the vertex is shifted right. If h is negative, then the vertex is shifted left.

right and a positive sign indicates a translation to the left. The key is the minus sign in $g(x) = |x - h| + k$. For a translation 3 units to the left, $h = -3$ and $k = 0$ in the general form $g(x) = |x - h| + k$. In that case, the function is written $g(x) = |x - (-3)| = |x + 3|$.

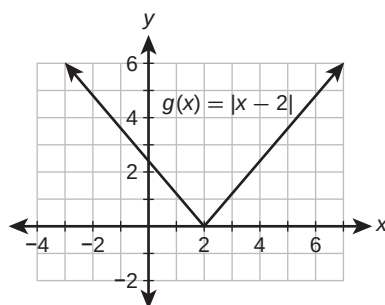
EXAMPLE 2

Graph the function.

A $g(x) = |x - 2|$

SOLUTION

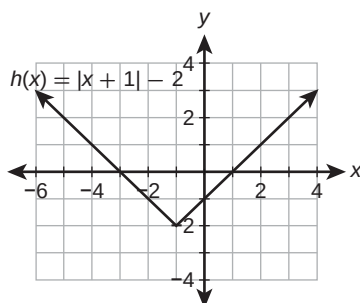
The graph is a horizontal translation of the parent function, 2 units to the right. You can write the function as $g(x) = |x - 2| + 0$; the vertex (h, k) is located at $(2, 0)$.



B $h(x) = |x + 1| - 2$

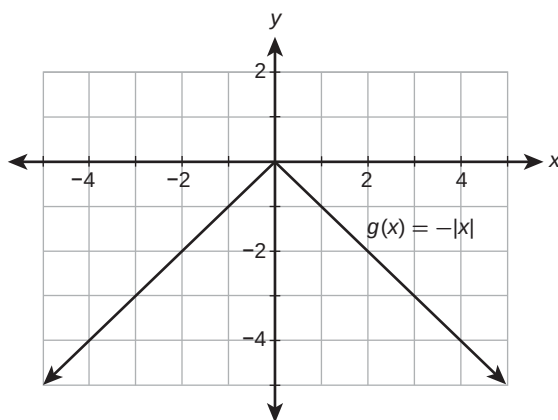
SOLUTION

The graph is a vertical and horizontal translation of the parent function. It is translated 1 unit left and 2 units down. You can write the function as $h(x) = |x - (-1)| + (-2)$; the vertex is located at $(-1, -2)$.



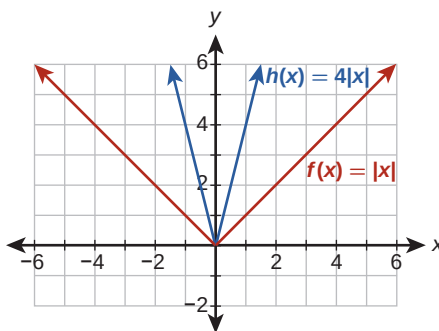
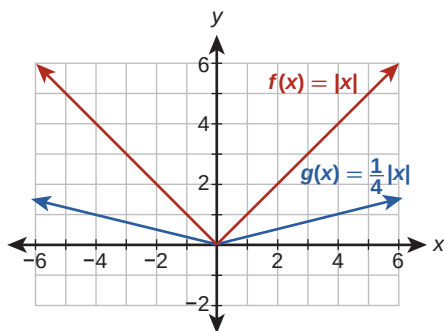
Describing Graphs of Absolute Value Functions

When the absolute value expression in the parent function $f(x) = |x|$ is multiplied by -1 , the graph is reflected across the x -axis, making an upside-down V-shape. You can write this function as $g(x) = -1|x|$ or $g(x) = -|x|$.



If the absolute value expression is multiplied by any factor other than 1 or -1 , the graph becomes narrower or wider. For $g(x) = a|x|$, the graph becomes wider (or vertically compressed) if $0 < |a| < 1$ and narrower (or vertically stretched) if $|a| > 1$.

Compare the graphs of $g(x) = \frac{1}{4}|x|$ and $h(x) = 4|x|$ to $f(x) = |x|$.



EXAMPLE 3

Describe how the graph of the function differs from the graph of $f(x) = |x|$. Then determine the domain and range.

A $g(x) = 0.8|x|$

SOLUTION

Because $|a|$ is between 0 and 1, the graph will be wider than the graph of the parent function. The vertex remains at $(0, 0)$.

domain: $\{x \mid x \in \mathbb{R}\}$, range: $\{y \mid y \geq 0\}$

B $g(x) = 3|x - 8|$

SOLUTION

The graph is translated 8 units to the right. Because $|a| > 1$, the graph will be narrower than the graph of the parent function.

domain: $\{x \mid x \in \mathbb{R}\}$, range: $\{y \mid y \geq 0\}$

C $g(x) = -2|x| + 5$

SOLUTION

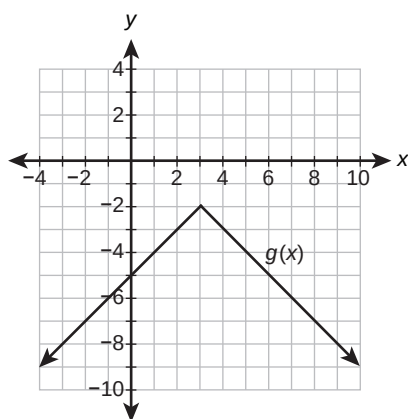
The graph is translated 5 units up. Because $|a| > 1$, the graph will be narrower than the graph of the parent function. Because a is negative, the shape of the graph is an upside-down V.

domain: $\{x \mid x \in \mathbb{R}\}$, range: $\{y \mid y \leq 5\}$ ■

Finding the Equation of an Absolute Value Function

EXAMPLE 4

Write the equation of $g(x)$, which is graphed.



SOLUTION

The vertex is located at $(3, -2)$. Using the parent function $f(x) = |x|$ as a reference, the horizontal and vertical translations indicate that $h = 3$ and $k = -2$.

The graph, which moves up or down 1 unit for every horizontal move of 1 unit, is neither narrower nor wider than that of the parent function, so $|a| = 1$. Because the graph is upside-down, $a = -1$, making the equation $g(x) = -|x - 3| - 2$. ■